

UNIT 5

PLANT LIFE CYCLES

Learning objectives:

- Identify roots, stems, leaves, flowers, fruits, and seeds and describe what they do.
- Explain how plants in rainforests, deserts, and other places change to survive.
- Understand that flowers make seeds, fruits protect seeds, and seeds grow into new plants.
- Explain how animals, wind, and water help move seeds to new places.
- Describe how a plant grows from a seed, changes, and makes new plants.
- Explain how young plants look like their parent plants.
- Understand that plants need light, water, and soil to grow

Plants and Their Parts

● What Do Roots, Stems, and Leaves Do?

Plants use their roots, stems, and leaves to get light and water.

Leaves use light to make food.



The stem holds Up the plant.

Roots hold the plant in the ground. They take in water from the soil

Plants that grow in the rain forest have large leaves. Large leaves help plants take in sunlight.

Desert plants have few leaves. Some desert plants have no leaves. They store water in their thick stems.



Banana Tree



Joshua Tree

- **How Can We Describe Roots?**

Some roots are long and thin. Others are short and thick. Plants that live in dry places may have long roots. They grow down to find water in the ground. Plants that live in wet places may have roots above the ground. That way, the plant does not get too much water.



▲ **The banyan tree lives in a wet place. Its roots grow above the ground.**



Roots also help animals. Bears, raccoons, and porcupines are some of the animals that eat roots.

We eat roots, too. Radishes, carrots, and beets are some of the roots we eat.



Carrots



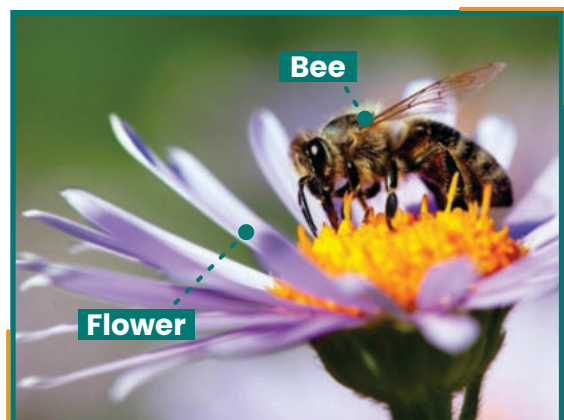
Beetroot

• What Do Flowers, Fruit, and Seeds Do?

Many plants have flowers. **Flowers** are the parts of plants that make seeds. Inside a flower is a sticky powder called **pollen**. Pollen helps flowers make seeds.



▲ We eat flowers of the broccoli plant.



▲ A bee can move pollen from one flower to another



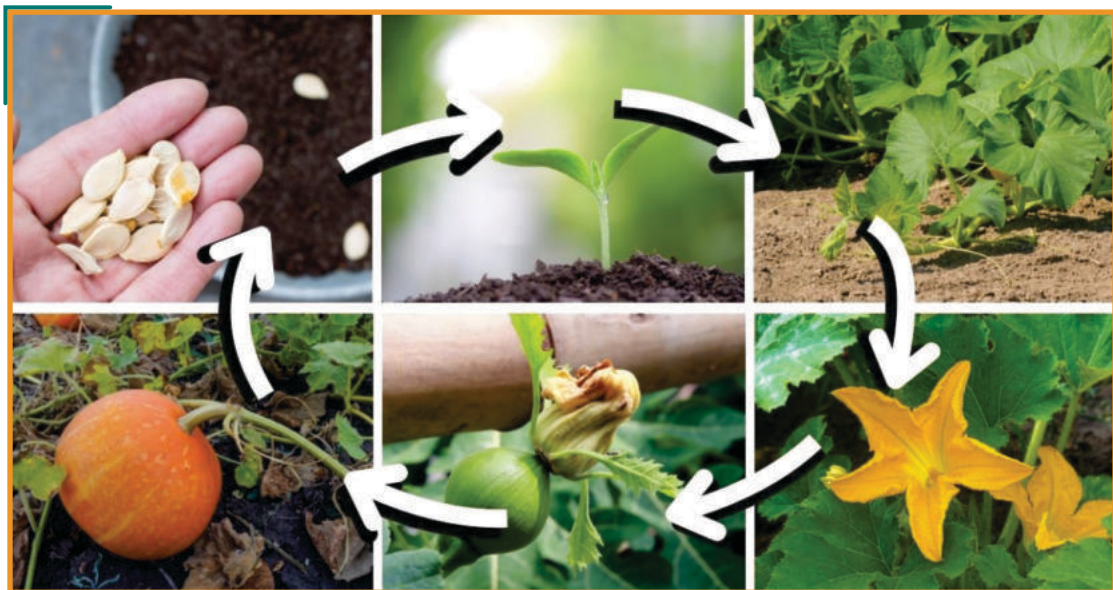
Plants that have flowers make fruit. Most of the time, seeds grow inside a fruit. The **fruit** keeps the seeds safe and helps them grow. The **seeds** can grow into new plants. We eat the fruits of many plants, such as apples.



Flowers and Fruits

• How Do Flowers Make Seeds?

One part of the flower makes pollen. Another part of the flower uses the pollen to make seeds. The seeds can grow into new plants.



• How Do Seeds Look?

Seeds have many different shapes and sizes. They all need light, water, and food to grow.

Marigold Seeds

Marigold seeds are small and thin. ▶



Marigold

They all need light,
water, and food
to grow.

Star Anise



Star anise pod

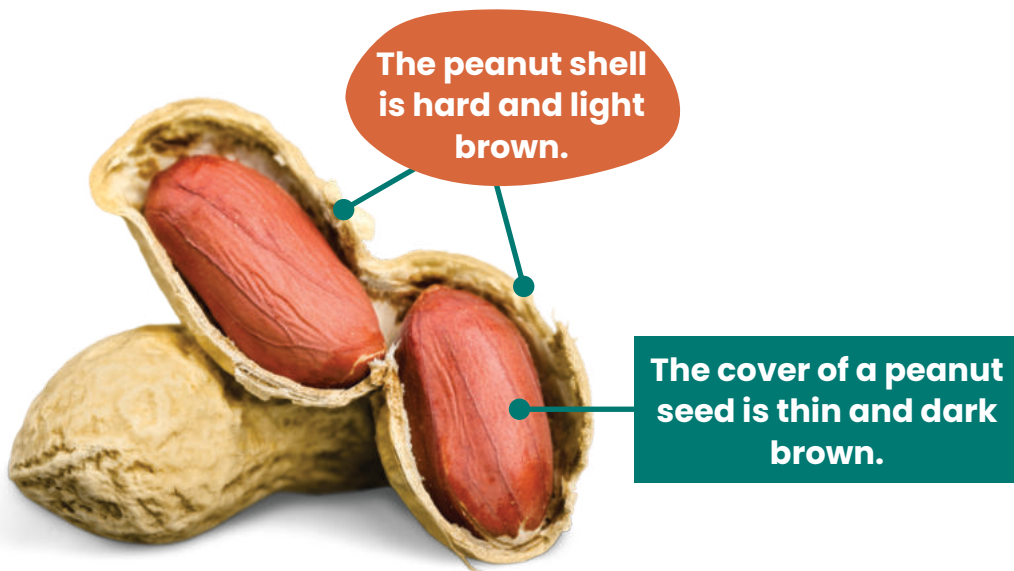
▶ A pod is a shell for seeds. This pod has room for many seeds.



All seeds have covers that protect them. The covers also help keep the seeds from drying out. Some seeds, like peanuts, also have shells.



▲ **Peanuts are seeds.**
They grow underground.



● **How Do Seeds Move?**

Animals help move seeds to new places. Many animals eat fruit and leave the seeds behind. Some animals, like squirrels, bury seeds. Some seeds stick to an animal's fur and move to new places.



▲ **This baboon eats fruit with seeds.**



▲ **This carries seeds on its fur.**

Oceans and rivers can move seeds, too. Seeds fall into the water and are carried to new places. Wind can also carry seeds far away.



▲ **The wind can carry this maple tree seed to a different place.**



Plants Grow and Change

- **How Are Plants Like Their Parents?**

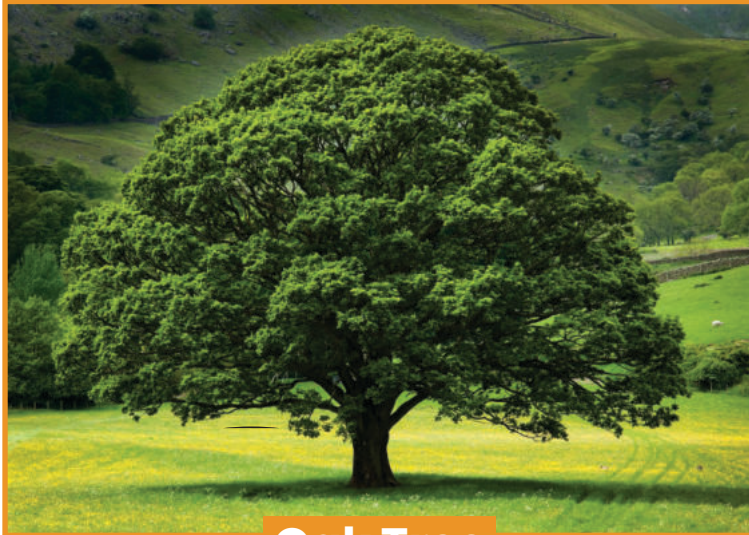
You know that animals have babies that look and act like their parents. Plants are the same way.

A sunflower makes seeds that grow into sunflowers. An oak tree makes acorns that grow into oak trees.

Sunflower



Sunflower seeds



Oak Tree



acorn



Most young plants look like their parent plants. Their flowers, petals, and leaves will have the same shape. Some plants, like tulips, may look a little different from their parent plants.



▲ Tulips come in different colors.

• What Is a Life Cycle?

A **life cycle** shows how a living thing grows, changes, and makes new living things. The plant life cycle begins with a seed. It keeps going as plants make new plants.



Plants follow the same life cycles as their parent plants. Different plants have different life cycles. Some plants live for a few weeks. Others live for many years.



▲ These flowers go through their life cycle in a few months.



▲ Redwood trees take more than two years to make cones.

Plants and Their Homes

• How Can Plants Change to Get What They Need?

Plants need light to grow. Their parts can move to get the light they need. The stems and leaves can bend toward the light. Flowers can turn toward the light, too.



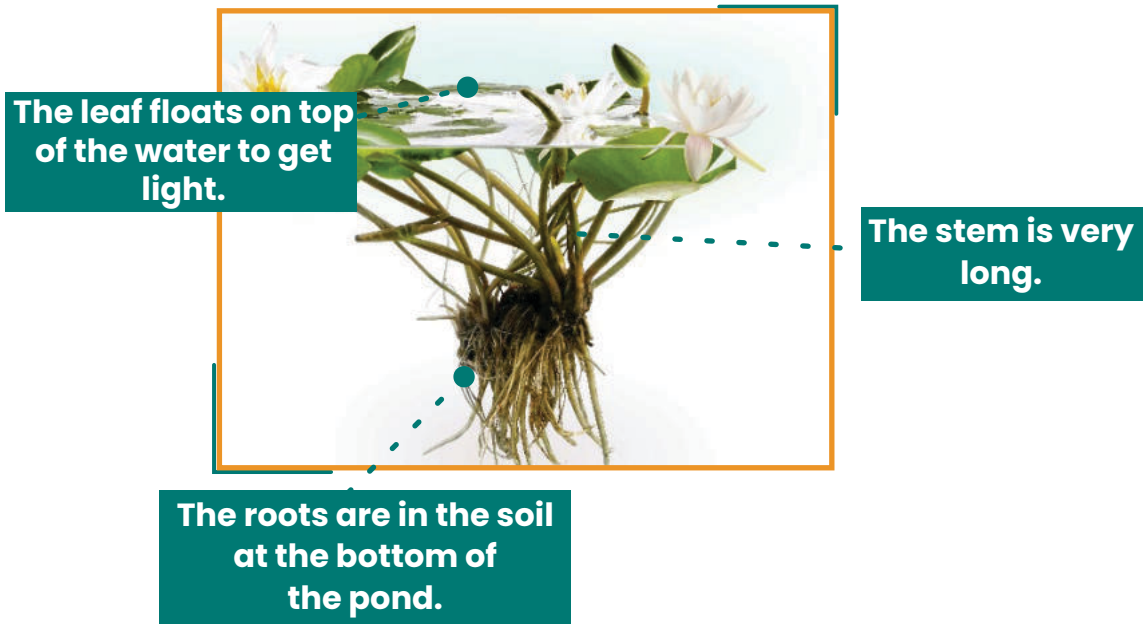
▲ This plant bends toward the light that comes through the window.



▲ Some flower turn to face the sun as it moves across the sky.



Plants need soil to grow. They take in nutrients and water from the soil. The roots of a water lily grow down in the soil. The stem grows up toward the light.



• What Helps Plants Live in Their Homes?

Plants have different ways to stay safe and get what they need. The willow tree has long roots to get water from deep in the ground. The mimosa plant closes its leaves when it is touched. This keeps animals from eating them.



▲ The leaves of this mimosa plant are open.



▲ The leaves mimosa plant are closed.



The weather causes some plants to change the way they grow.



▲ **This tree growing to one side because of strong winds.**



Test Your Understanding

Q1: What do the roots of a plant do?

Ans: _____

Q2: What does a flower grow into after its petals fall off?

Ans: _____

Q3: What is the function of a fruit in a plant?

Ans: _____

Q4: How do seeds move from one place to another?

Ans: _____



Q5: Why do plants need light?

Ans: _____

Q6: What do sunflower seeds grow up to be?

Ans: _____

Q7: How are young plants like their parents?

Ans: _____

Q8: What is a life cycle?

Ans: _____



Fill in the blanks.

1. _____ are the parts of plants that make seeds.
2. Pollen is a sticky powder inside _____.
3. The _____ protects the seeds and helps them grow.
4. Plants need _____, water, and food to grow.
5. Animals like squirrels help move seeds by _____ them.
6. The life cycle of a plant begins with a _____.
7. Plants that live in dry places may have _____ roots.
8. We eat roots such as carrots and _____.
9. A fruit's seeds can grow into new _____
10. A mimosa plant _____ its leaves when it is touched.



Tick (✓) for True and cross (x) for False statements.

1. Seeds do not need light to grow. ()

2. The roots of a water lily grow above the ground. ()

3. Some plants, like tulips, may look a little ()

different from their parent plants.

4. Pollen helps flowers make seeds. ()

5. The stem of a plant holds it up and transports ()

water.

6. All plants have the same life cycle. ()



Choose the correct answer:

1. What part of the plant holds it in the ground?

- a) Stem
- b) Leaves
- c) Roots
- d) Flowers

2. Which of the following helps plants make food?

- a) Roots
- b) Stems
- c) Flowers
- d) Leaves

3. Which of the following moves pollen from one flower to another?

- a) Rocks
- b) Animals
- c) Soil
- d) Sunlight

4. What protects the seeds inside a plant?

- a) Roots
- b) Stems
- c) Fruit
- d) Leaves





5. Which of the following shows how a living thing grows and changes?

- a) Root system
- b) Flower cycle
- c) Life cycle
- d) Seed coat

6. What part of the plant makes pollen?

- a) Roots
- b) Leaves
- c) Flowers
- d) Stems



Match the Column

Flowers

Sticky powder inside flowers

Life cycle

Parts of plants that make seeds

Seeds

Keeps seeds safe and helps them grow

Fruit

Can grow into new plants

Pollen

Steps that show how a living thing grows





Activity 1: Seed Dispersal Experiment



ACTIVITY

Collect different seeds (e.g., dandelion, maple, sunflower) and test how they move. Drop them from a height and observe if they float, fall straight down, or stick to objects. Record which seeds travel the farthest and discuss why certain seeds are adapted to specific environments.



Activity 2: Create a Plant Life Cycle Diagram



ACTIVITY

Using paper and colored pencils, draw the life cycle of a plant starting from a seed, showing each stage—seed, sprout, young plant, mature plant, and flower. Label each part and describe what happens in each stage.





Activity 3: Observation Journal



ACTIVITY

Plant a seed in a small pot and observe its growth over a few weeks. Record the changes in height, the appearance of leaves, and the development of flowers or fruits. Take pictures or draw diagrams of the changes you observe.



Activity 4: Leaf Light Experiment:



ACTIVITY

Take two similar plants and place one in sunlight and the other in a shaded area. Observe and record how the leaves react after a few days. Discuss the importance of light in photosynthesis and plant growth.





Activity 5: Root Growth Exploration



ACTIVITY

Grow a bean plant in a clear plastic cup with wet paper towels. Observe how the roots grow downward and the stem grows upward. Draw and label the parts of the plant.



LIFECYCLE OF A PINE TREE

Complete the life cycle of a pine tree.

